

Incremental accelerated gradient descent and adaptive fine-tuning heuristic performance optimization for robotic motion planning

Shengjie Li^{a,b}, Jin Wang^{a,b,*}, Haiyun Zhang^{c,d}, Yichang Feng^{a,b}, Guodong Lu^{a,b}, Anbang Zhai^{a,b}

^a State Key Laboratory of Fluid Power and Mechatronic Systems, School of Mechanical Engineering, Zhejiang University, Hangzhou 310027, China

^b Robotics Institute of Zhejiang University, Hangzhou 310027, China

^c School of Robotics, Ningbo University of Technology, Ningbo 315211, China

^d Robotics Research Center of Yuyao City, Ningbo 315400, China

ARTICLE INFO

Keywords:

Motion planning
Performance evaluation
Heuristic optimization
Accelerated gradient descent
Incremental optimization

ABSTRACT

Robot comprehensive performance optimization is a high dimensional non-convex problem, which is of great significance in the practice of robotic intelligent manufacturing. Existing methods still face many challenges in performance evaluation, local optima, and planning efficiency. In this paper, a novel method based on accelerated gradient descent and directed-mutagenesis genetic algorithm (AGD-DGA) is proposed for efficient robotic motion planning and comprehensive performance optimization. Firstly, the time, energy consumption, and impact (T-E-I) are taken as the optimization objectives. In particular, the power loss sub-model is constructed, which is rarely considered in the previous research on motion optimization, and it is combined with the driving power sub-model to improve the evaluation accuracy. Next, a novel directed-mutagenesis genetic algorithm (DGA) is proposed to address the local optima problem. Compared with existing methods based on random sampling, the DGA further improves the success rate and efficiency by updating the probability model based on the adaptive Gaussian process. Moreover, a hierarchical weight updating mechanism is established to overcome the impact of additional performance indicators on the success rate of collision-free path planning. To improve planning efficiency, an incremental optimization mechanism and an adaptive fine-tuning strategy for population size are proposed to reduce unnecessary sampling, evaluation, and sorting. The simulation and experimental results in different scenarios demonstrate that the proposed AGD-DGA achieves better results than most of the mainstream algorithms in terms of optimization efficiency, success rate, and motion performance. In addition, it is also verified that the modified energy consumption model can achieve an average evaluation accuracy of more than 90% while maintaining the calculation efficiency.

1. Introduction

In robotic intelligent manufacturing applications, motion time, energy consumption, and motion impact are important factors for evaluating robotic comprehensive working performance. For example, in the automotive industry, robots consume 8 % of the total energy used in the production phase (Paryanto, Kohl, Merhof, Spreng, & Franke, 2014) and emit approximately 50 tons of CO₂ for each vehicle (Paulus, 2014). Considering the tasks with high repeatability, high load, and high complexity in industrial scenarios, such as stacking, handling, and liquid transportation as shown in Fig. 1, motion optimization can bring about

significant gains in efficiency, energy, and impact suppression (Kayastha, Katupitiya, & Pearce, 2023; Pastras, Fysikopoulos, & Chrysosolouris, 2019).

The methods based on the gradient descent technique are widely used because of their ability to achieve stable and efficient convergence. Zucker et al., (Zucker et al., 2013) proposed Covariant Hamiltonian optimization for motion planning (CHOMP), which used functional gradient techniques for collision-free smooth trajectory optimization, and used Hamiltonian Monte Carlo to alleviate the problem of convergence to high-cost local minima. Kang et al., (Kang, Shin, Kim, & Yoon, 2020) adopted a two-stage gradient descent to reduce potential

* Corresponding author.

E-mail addresses: 11825057@zju.edu.cn (S. Li), dwjcom@zju.edu.cn (J. Wang), gray_sun@zju.edu.cn (H. Zhang), yichangfeng@zju.edu.cn (Y. Feng), lugd@zju.edu.cn (G. Lu), abzhai@zju.edu.cn (A. Zhai).

<https://doi.org/10.1016/j.eswa.2023.122794>

Received 25 August 2023; Received in revised form 15 November 2023; Accepted 28 November 2023

Available online 29 November 2023

0957-4174/© 2023 Elsevier Ltd. All rights reserved.

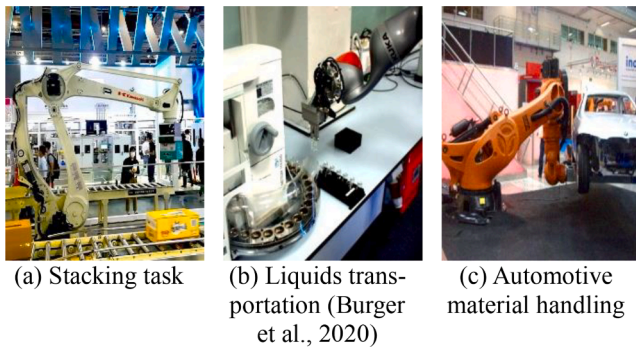


Fig. 1. Some scenarios that require performance optimization. (See above-mentioned references for further information.)

competition between different properties during the update. Safeea et al., (Safeea, Béarée, & Neto, 2020) utilized the artificial potential field and Newton method to plan collision-free motion for a 15-DOF manipulator, eliminating oscillation and accelerating convergence. However, the original gradient-based optimization technique is prone to local optima in narrow feasible regions, which reduces the success rate of planning. Therefore, stochastic gradient descent has been adopted to avoid local optimization and improve convergence efficiency (Henkel & Toussaint, 2020). Feng et al., (Feng, Wang, Zhang, & Lu, 2022) proposed an incremental stochastic and accelerated gradient descent optimization method (ISAGO) based on Bayesian tree inference and achieved a higher planning success rate. Moreover, Motion planning algorithms based on gradient descent and stochastic processes, such as GPMP2 (Mukadam, Dong, Yan, Dellaert, & Boots, 2018), have attracted attention. GPMP2 utilizes the incremental smoothing and mapping algorithm (iSAM) and Bayes tree inference (BTI) to perform nonlinear incremental optimization for high-dimensional problems. In recent studies, GPMP2 has been widely applied to SLAM and motion planning because it can solve sparse problems with fewer computational resources (Dong, Mukadam, Boots, & Dellaert, 2018).

Stochastic optimization also plays a crucial role in motion optimization, especially in addressing the issue of local minimum. Osa et al., (Osa, 2020) proposed a stochastic multimodal trajectory optimization algorithm, which transformed the optimization problem into a density estimation problem among multiple candidate solutions. Moreover, the asymptotic optimal algorithms based on RRT* have also received attention (Noreen, Khan, Asghar, & Habib, 2019; Yi, Yuan, Sun, & Bai, 2022). It should be noted that although the stochastic optimization methods usually have probabilistic completeness and can effectively avoid local optima, their time complexity is often significantly higher than that of gradient-based methods.

Motion optimization based on heuristic algorithms is also considered to be a promising technique. In recent years, several motion planning approaches based on genetic algorithm (Yu, Dong, & Yin, 2022), particle swarm optimization (Ekrem & Aksoy, 2023; Yang, Xu, Li, Hua, Zhang, Ling, Yao, & Ming, 2022; Zhang, Zhang, Zhang, & Wei, 2022), gray wolf optimization (Du, Yu, & Chen, 2023) and so on have emerged. Moreover, some studies have shown that hybrid heuristic algorithms have the potential to further improve success rate and efficiency (Singh & Banga, 2022). However, in terms of efficiency, they are still inferior to gradient-based algorithms.

Besides optimization methods, performance evaluation is another crucial aspect of performance optimization. Most existing energy consumption evaluation methods are based on the “square of torque” (Li, Wang, Yang, Zhang, & Wei, 2023), “square of acceleration” (Cao et al., 2021; Liu, Wang, Li, Chen, & Li, 2023), and “dynamics model” (Ghorbanpour & Richter, 2022). These methods are computationally efficient and can be used for multiple calls in performance optimization. However, the accuracy of these simplified methods is insufficient, which may

affect the validity of optimization results. To evaluate motion impact, the joint impact is widely used as the evaluation index. Fang et al., (Fang & Ding, 2022) adopted the third derivative of joint angular displacement as a constraint to realize sampling-based motion planning. Zhang et al. (Zhang, Zhang, & Zou, 2021) used joint torque to represent the joint impact and smoothed the trajectory with a cubic spline curve on the phase plane.

According to the above literature, there are still several important issues in robotic collision-free motion planning and performance optimization remain to be further studied:

- (1) **Performance evaluation.** Robotic comprehensive performance optimization still faces numerous challenges. To date, few studies have considered the simultaneous optimization of T-E-I performance indicators along with collision constraints. Current energy consumption models, such as the “square of torque” (Li et al., 2023) or “dynamics model” (Ghorbanpour & Richter, 2022), cannot accurately evaluate the energy loss when the joints are in the stationary state, which reduces the validity of the energy consumption optimization results. Meanwhile, most of the current impact models focus on the smoothness of joint motion. However, for some tasks such as handling flexible parts, liquids, or hazardous materials, the motion impact of the end-effector may be more important, which is not always positively correlated with the joint impact (e.g. when approaching singular configurations). Therefore, the end motion impact deserves further analysis.
- (2) **Success rate.** In the high-dimensional solution spaces, especially in the case of narrow feasible space, the gradient descent algorithm and its variants (Safeea et al., 2020; Zucker et al., 2013) face local optimal problems. Although the traditional random sampling approaches have probabilistic completeness, their computational complexity is sensitive to state dimension (Zhong, Wang, & Cheng, 2022), and it is difficult to achieve a satisfactory success rate under limited iterations. Moreover, when the feasible space is narrow, the additional performance indices will affect the success rate, even if they are significantly less weighted than soft constraints (collision, joint limits, etc.).
- (3) **Planning efficiency.** Combining the gradient descent technique with stochastic optimization is a mainstream research approach at present (Feng et al., 2022; Henkel & Toussaint, 2020; Mukadam et al., 2018; Osa, 2020), and its planning efficiency is affected by many factors. Gradient descent trajectory optimization requires repeated upsampling, which significantly increases the time complexity of the whole trajectory optimization. Moreover, the methods based on heuristics or stochastic algorithms require manual setting of the sampling scale, which is not always effective, improper sampling scale will lead to planning failures or lower efficiency.

To address the above-mentioned issues, this paper proposes an accelerated gradient descent and directed-mutagenesis genetic algorithm (AGD-DGA) for efficient robotic motion planning and comprehensive performance optimization. The main contributions are as follows:

- (1) Considering the comprehensive performance evaluation problem, a driving-loss model (DLM) is proposed to evaluate energy consumption, and a jerk-based model is adopted to evaluate end impact. The DLM models driving power and power loss, the latter of which is neglected by mainstream energy consumption optimization methods. The jerk of the end-effector motion is calculated based on the forward kinematics and central difference. In addition to the above performance models, the basic constraints, such as collision and joint limits, are transformed into penalty functions and integrated into the objective function.

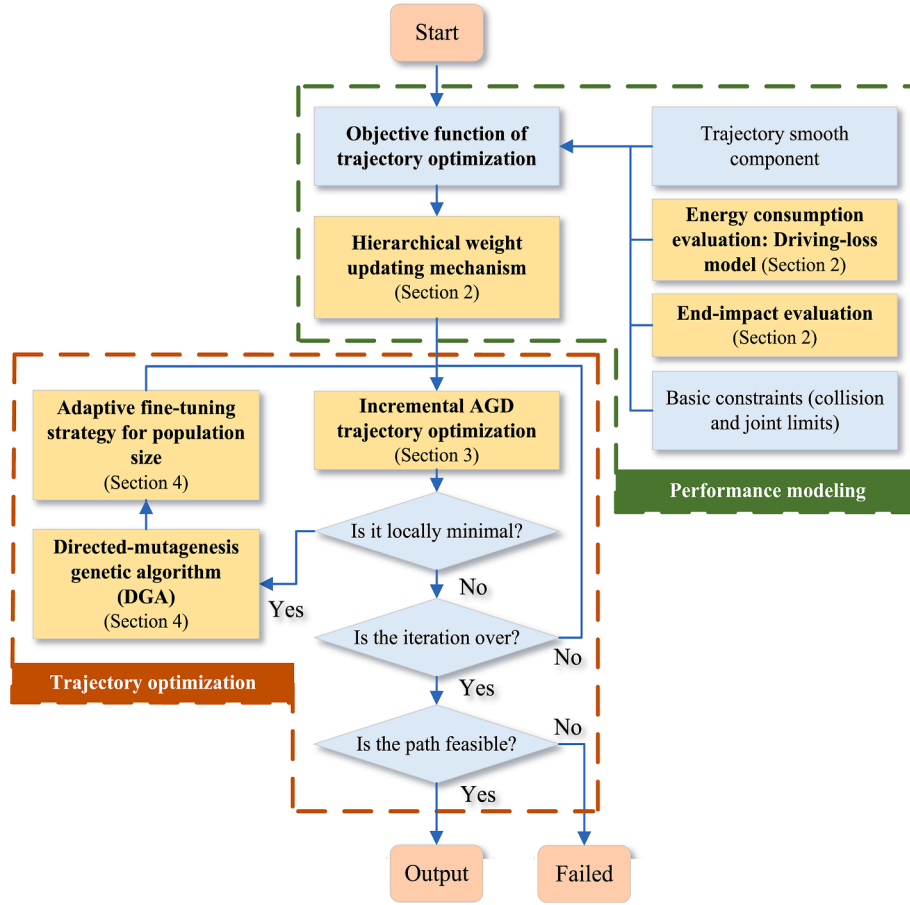


Fig. 2. The framework of the proposed AGD-DGA.

Table 1
Acronyms and meanings.

Acronym	Meaning
T-E-I	Time, Energy consumption and Impact
SDF	Signed Distance Field
CHOMP	Covariant Hamiltonian Optimization for Motion Planning
STOMP	Stochastic Trajectory Optimization for Motion Planning
AGD	Accelerated Gradient Descent
iSAGO	Incremental Stochastic and Accelerated Gradient information mixed Optimization
AFS-PS	Adaptive Fine-tuning Strategy for Population Size
AGD-DGA	Accelerated Gradient Descent and Directed- mutagenesis Genetic Algorithm
DLM	Driving-Loss Model
GA	Genetic Algorithm
AGP	Adaptive Gaussian Process
DGA	Directed-mutagenesis Genetic Algorithm
IRLS	Iteratively Reweighted Least Squares
HWU	Hierarchical Weight Updating mechanism

- (2) To improve the success rate of planning in a narrow space, a hierarchical weight update mechanism is established, which dynamically adjusts the weight of additional performance indicators to ensure the feasibility of the trajectory. Moreover, a novel directed-mutagenesis genetic algorithm is proposed to address the difficulty of effective convergence in traditional stochastic optimization methods. The expectation maximization (Do & Batzoglu, 2008) and the exponential moving average (Mak, 2021) are adopted to update the probability model, enabling the probability model to be adjusted adaptively according to the optimization objective. Then the adaptive Gaussian process is

introduced into the genetic algorithm and a directed mutagenesis operation is proposed. The original mutation operation of the original genetic algorithm is retained to keep the global exploration ability in theory.

- (3) To improve the motion planning efficiency, an incremental optimization mechanism is established for Accelerated Gradient Descent (AGD) that accelerates the process by targeting high-cost path points. An adaptive fine-tuning strategy is also proposed for DGA that dynamically adjusts the population size according to the cost gradient of adjacent iterations, saving unnecessary sampling, evaluation, and sorting.

Fig. 2 shows the framework of the proposed AGD-DGA, and Table 1 lists the acronyms adopted in this article and their meanings.

The rest of this paper is organized as follows. Section 2 formulates the objective function for the motion performance indices, especially energy consumption, and proposes an improved model DLM. Section 3 proposes the incremental optimization mechanism for AGD and upsampling. Section 4 proposes an adaptive fine-tuning strategy and DGA to address the issues of efficiency and success rate. Section 5 presents and analyzes the simulation and experimental results. Finally, Section 6 concludes this study.

2. Performance modeling and objective function construction

To improve the motion performance of the manipulator, this study considers the performance indices of time, energy, and impact. Generally, the objective function of trajectory optimization can be obtained as Eq. (1):

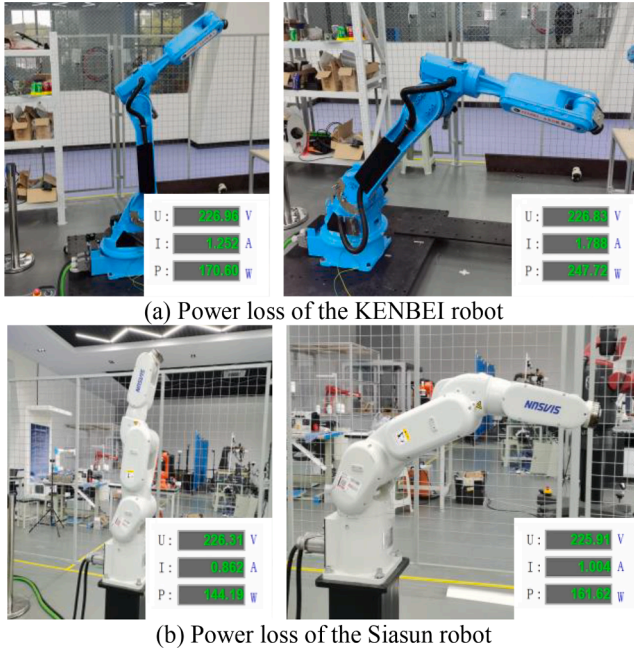


Fig. 3. Power loss at stationary of different configurations.

$$F(\theta) = w_1 f_{gp}(\theta) + w_2 f_E(\theta) + w_3 f_i(\theta) + w_4 f_{obs}(\theta) + w_5 f_L(\theta) \quad (1)$$

where f_{gp} forms the smooth component (the specific calculation method can be referred to (Feng et al., 2022)), f_E is the energy consumption cost, f_i is the impact cost, f_{obs} is the obstacle cost, f_L is used to prevent the angular displacement from exceeding limits, and w_1 to w_5 are the weights of the costs. The modeling methods for energy consumption and impact indicators are described below.

2.1. Motion performance modeling

Energy consumption evaluation primarily consists of motion energy and stationary loss. Given the motion states of each joint (i.e. angle, velocity, and acceleration), the torque of each joint is calculated based on the Newton-Euler method:

$$\tau_{dyn,i} = M(\theta_i)\ddot{\theta}_i + V(\theta_i, \dot{\theta}_i) + G(\theta_i) + \tau_{f,i}(\dot{\theta}_i) \quad (2)$$

where $M(\theta_i)$ is the inertial matrix, $V(\theta_i, \dot{\theta}_i)$ is the vector containing the Coriolis and centripetal force moment, and $G(\theta_i)$ is the vector containing the gravity moment. In particular, the friction torque $\tau_{f,i}$ of each joint can be calculated based on the Karnopp friction model (Karnopp, 1985), which can reflect the Stribeck effect well. In the Karnopp model, the friction force depends solely on the current state, as shown in Eq. (3).

$$\tau_{f,i}(\dot{\theta}_i) = \begin{cases} \text{sgn}(\dot{\theta}_i) \cdot (f_{c,i} + (f_{s,i} - f_{c,i})e^r) + f_{v,i}\dot{\theta}_i & |\dot{\theta}_i| > DV \\ f_{stick,i} & |\dot{\theta}_i| \leq DV \end{cases} \quad (3)$$

where $f_{c,i}$ is the Coulomb friction, $f_{s,i}$ is the static friction, $f_{v,i}$ is the coefficient of viscous friction, DV is a velocity threshold approaching zero, while $f_{stick,i}$ and r are defined in Eqs. (4) and (5), respectively:

$$f_{stick,i} = \begin{cases} f_{e,i} & |f_{e,i}| < f_{s,i} \\ f_{s,i} \cdot \text{sgn}(f_{e,i}) & |f_{e,i}| \geq f_{s,i} \end{cases} \quad (4)$$

$$r = - \left(\frac{\dot{\theta}_i}{\dot{\theta}_{s,i}} \right)^2 \quad (5)$$

where $f_{e,i}$ is the external friction, and $\dot{\theta}_{s,i}$ is the Stribeck velocity. The friction torque measurement and parameter identification method can be found in the literature (B. Zhang, Wang, & Wu, 2019), and will not be repeated here.

Based on the above calculation results, the driving power of the i^{th} joint of the manipulator can be estimated according to the following equation:

$$P_{d,i} = \frac{\tau_{dyn,i}\dot{\theta}_i}{\eta_{m,i} \cdot \eta_{r,i}} \quad (6)$$

where $\eta_{m,i}$ is the efficiency of the i^{th} joint's motor, and $\eta_{r,i}$ is the efficiency of the i^{th} joint's reducer.

Moreover, the power loss at stationary during power-on (denoted as P_l) is also considered in this study. Generally, the joint torque and velocity are the main factors affecting power (Paryanto, Kohl, Merhof, Spreng, & Franke, 2014; Rassölnkin, Hõimoja, & Teemets, 2011). However, while the robot performs tasks, some joints may be stationary at certain time intervals ($\dot{\theta}_i = 0$), and the power evaluation results of these joints will be 0 according to Eq. (6), even though the joint motors and other components actually still consume energy. This difference can lead to significant deviations in the evaluation results. Moreover, the robotic pose configuration also affects the power loss as shown in Fig. 3.

Another challenge is that different robot systems have different controllers, motors, actuators and other electrical components, which makes it difficult to evaluate power loss using the existing energy consumption models. Therefore, to improve the evaluation accuracy, the power loss of multiple configurations is measured, and a fuzzy expert system for power loss evaluation is constructed. The expert system stores the regression models of power loss in its knowledge base. These models can be adopted based on the information of different robots, such as the type, rated load, and working range. Considering the fitting effect and the workload of data collection, the regression model is constructed based on IRLS in the form of a six-element cubic polynomial:

$$P_l = f(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6) = \sum_{i,j,k_1,k_2,k_3,k_4,k_5,k_6} a_i \cdot \theta_1^{k_1} \cdot \theta_2^{k_2} \cdot \theta_3^{k_3} \cdot \theta_4^{k_4} \cdot \theta_5^{k_5} \cdot \theta_6^{k_6} \quad (7)$$

where k_1 to k_6 are non-negative integers, the sum of k_1 to k_6 is an integer belonging to $[0,3]$, and a_i is the coefficient of the i^{th} term.

In summary, the comprehensive energy consumption of the entire process can be estimated according to Eq. (8):

$$f_E(\theta) = \int_{t_0}^{t_1} P_l dt + \sum_{i=1}^n \int_{t_0}^{t_1} P_{d,i} dt \quad (8)$$

where n is the number of joints, and the motion starts at time t_0 and ends at time t_1 . For convenience, this model is called DLM in this study.

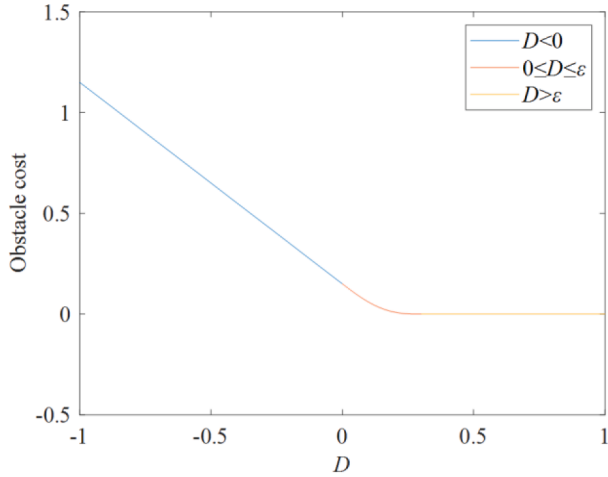
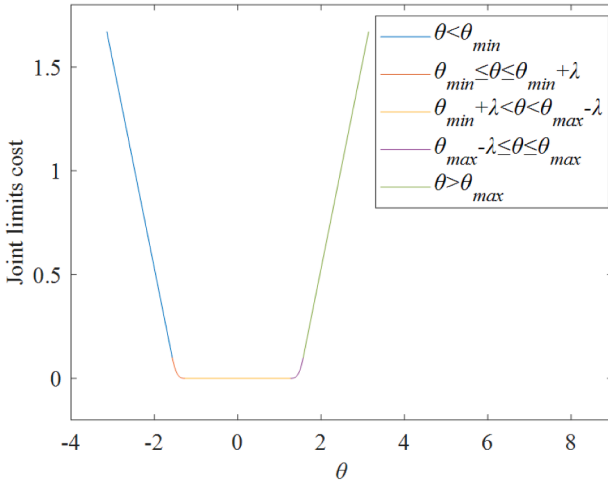
The estimation of impact is mainly based on the jerk of the end-effector in the movement process. Specifically, the integral value of jerk in the entire process is taken as the evaluation index of impact:

$$f_i(\theta) = \sum_{i=1}^n \int_{t_0}^{t_1} \left| \frac{\ddot{r}_1^i + \ddot{r}_2^i + \ddot{r}_3^i}{\pi} + \ddot{p}_1^i + \ddot{p}_2^i + \ddot{p}_3^i \right| dt \quad (9)$$

where r_1, r_2 , and r_3 are the attitude angles of the end-effector, p_1, p_2 , and p_3 are the displacements of the end-effector along the X, Y, and Z directions, respectively and n is the number of robot joints.

2.2. Motion constraints modeling

Collision-free trajectory optimization must also consider obstacle cost and joint angle constraints. To quantify the obstacle cost, the sphere chain of the manipulator and the SDF of obstacles are constructed. The SDF generation method in this study is the same as that of (Mukadam et al., 2018), so it is not described here. To keep the robot away from

(a) The curve of obstacle cost, where $\varepsilon = 0.3$.(b) The curve of joint angle cost, where $\lambda = 0.3$, $\theta_{min} = -\pi/2$, and $\theta_{max} = \pi/2$.**Fig. 4.** Cost function curves of basic constraints.

collision, the obstacle cost function for each sphere of the robot is set as Eq. (10):

$$c(D, \varepsilon) = \begin{cases} -D + \frac{\varepsilon}{2} & \text{if } D < 0 \\ -\frac{1}{2\varepsilon^3}(-D + \varepsilon)^4 + \frac{1}{\varepsilon^2}(-D + \varepsilon)^3 & \text{if } 0 \leq D \leq \varepsilon \\ 0 & \text{if } D > \varepsilon \end{cases} \quad (10)$$

where ε is the sum of the sphere radius and the collision clearance (c_c), and D is the signed number of each point. The above cost function provides a continuous and smooth obstacle cost curve, as shown in Fig. 4 (a), which is conducive to the calculation of the gradient. Based on Eq. (10), the obstacle cost of the robot can be calculated as follows:

$$f_{obs}(\theta) = \sum_{j=1}^m c_j(D, \varepsilon_j) \|\dot{x}_j\| \quad (11)$$

where $\|\dot{x}_j\|$ is the norm of the velocity vector at the current moment of each sphere, j represents the serial number of each sphere, and m is the number of spheres. The cost of joint angle limitation is quantified similarly. To ensure that the constraint is satisfied, a margin λ is defined, and the cost is calculated as follows:

$$c_i^p(\theta_i, \lambda) = \begin{cases} \frac{\lambda}{3} + \theta_i^{min} - \theta_i & \text{if } \theta_i < \theta_i^{min} \\ \frac{(\theta_i^{min} - \theta_i + \lambda)^3}{3\lambda^2} & \text{if } \theta_i^{min} \leq \theta_i \leq \theta_i^{min} + \lambda \\ 0 & \text{if } \theta_i \in (\theta_i^{min} + \lambda, \theta_i^{max} - \lambda) \\ \frac{(\theta_i - \theta_i^{max} + \lambda)^3}{3\lambda^2} & \text{if } \theta_i^{max} \leq \theta_i \leq \theta_i^{max} - \lambda \\ \frac{\lambda}{3} + \theta_i - \theta_i^{max} & \text{if } \theta_i > \theta_i^{max} \end{cases} \quad (12)$$

where θ_i^{min} and θ_i^{max} are the lower and upper limits of the angle of the i^{th} joint, respectively. The term for the joint angle limitation is also continuous and differentiable, as shown in Fig. 4 (b), and the definition of f_L is shown in Eq. (13):

$$f_L(\theta) = \sum_{i=1}^n c_i^p(\theta_i, \lambda) \quad (13)$$

where n is the number of joints.

2.3. Hierarchical weight updating mechanism for motion performance

It is worth noting that adding the motion performance items, such as energy consumption and impact, can affect the success rate of planning in narrow spaces, even if their weights are significantly lower than that of soft constraint items. Meanwhile, reducing the weights of these indicators can also worsen the performance of optimization. Therefore, this study proposes a hierarchical weight updating mechanism (HWU) based on the priorities of different performance indicators, whose process is shown in Algorithm 1.

The HWU prioritizes smooth and soft constraint items over energy consumption and impact items. The entire optimization process is divided into two stages: stage I and stage II. In stage I, the main part of the optimization objective is prioritized. In stage II, the optimization is carried out according to the desired weight configuration. In the scenarios with sparse obstacles, due to the wide feasible space, reducing the weight of the second priority has little effect on the success rate, and even hurts the motion performance. Therefore, HWU also adjusts the weight update strategy according to the obstacle cost.

3. Incremental AGD-based motion optimization

In trajectory optimization, upsampling is an effective approach that can gradually adjust the trajectory based on a sparse set of support states, even if there are only starting and ending states at the beginning. Fig. 5 illustrates the trajectory optimization process based on upsampling, where P_0 indicates the start point and P_N indicates the goal point of the path. Upsampling can also be used for interpolation to directly output the final trajectory (Mukadam et al., 2018). However, with repeated upsampling operations, the calculation load of optimization will increase significantly. Controlling the number of path points is an intuitive solution. However, for long paths, dense path points are essential to ensure no collision. To improve the optimization efficiency, this study establishes an incremental optimization mechanism from the perspective of the necessity of the optimization of each path point.

First, the path points with high-cost values are filtered based on the probability metric (Kalakrishnan, Chitta, Theodorou, Pastor, & Schaal, 2011). The specific calculation process is shown in Eqs. (14) to (16).

$$w_F(\theta) = e^{-\frac{\lambda_p(\cos\theta - \cos\theta_{min})}{\cos\theta_{max} - \cos\theta_{min}}} \quad (14)$$

where λ_p is a constant for adjusting the probability distribution, while $\cos\theta_{min}$ and $\cos\theta_{max}$ are the minimum and maximum costs of the path points respectively. Note that the path points discussed here do not

Algorithm 1. Hierarchical weight updating mechanism (HWU).

Input: θ_0 , P_0 , P_N , N_{it} -the maximum number of iterations for upsampling, N_{io} -the maximum number of iterations for incremental optimization, N_p -the number of path points, N_{pmax} -the upper limit of N_p , N_{s1} -the number of iterations for stage 1, tol_{o1} -threshold of obstacle cost of θ , η_w -the coefficient of the second priority

```

1. Initialize: set  $w_{2r} = w_2$ ,  $w_{3r} = w_3$ ,  $\theta_0$ ;
2.  $\theta = \theta_0$ ;
3. for  $i = 1$  to  $N_{it}$  do
4.   if  $N_p < N_{pmax}$ 
5.     Upsample the trajectory  $\theta$ ;
6.   end
7.   for  $j = 1$  to  $N_{io}$  do
8.     if  $i \leq N_{s1}$  &  $f_{obs}(\theta) > tol_{o1}$  then
9.        $w_2 = w_{2r} \cdot \eta_w$ ;
10.       $w_3 = w_{3r} \cdot \eta_w$ ;
11.    else
12.       $w_2 = w_{2r}$ ;
13.       $w_3 = w_{3r}$ ;
14.    end
15.    Calculate the cost of each path point and update  $P_{hc}$ ;
16.    Optimize  $\theta$  based on incremental optimization;
17.  end
18. end

```

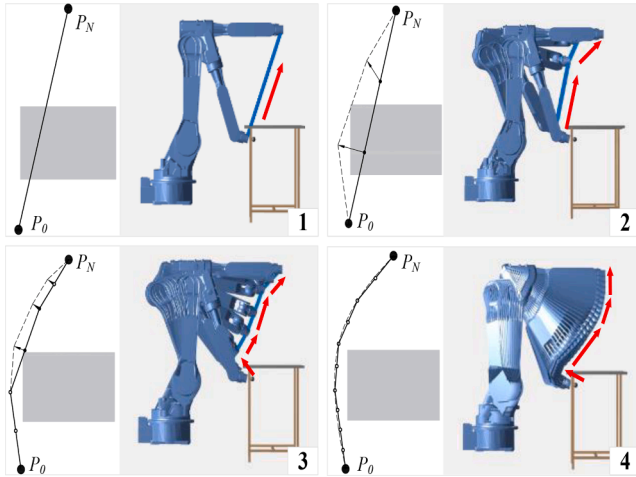


Fig. 5. Motion optimization based on upsampling.

include the start and goal points, which should remain unchanged during the optimization process. Thereafter, the probability metric of each path point can be calculated as follows:

$$W_{np} = \frac{W_{fp}}{\text{sum}(W_{fp})} \quad (15)$$

where W_{fp} is a vector containing the probability metrics of the path points. The set of local cost evaluation indices (LCEI) is defined as follows:

$$I_h = \text{round}((N_p - 2) \cdot W_{np}) \quad (16)$$

where N_p is the number of path points, and I_h contains the LCEI of each

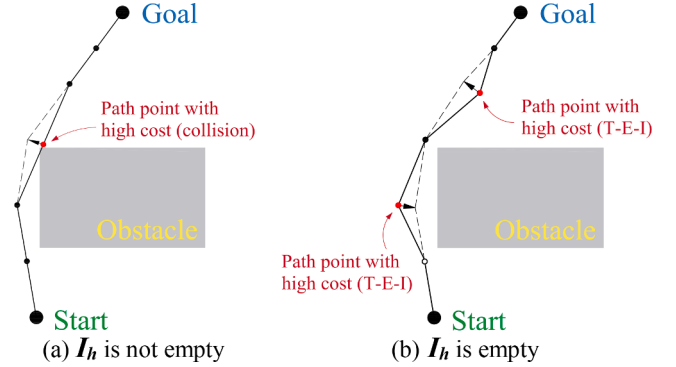


Fig. 6. The principle of incremental optimization.

path point. If the LCEI of some path point is greater than the preset threshold (set to 1 in this study), the point should be selected as a high-cost path point. The above screening mechanism effectively filters out the sub-trajectories that violate the basic constraints because the penalty functions for the basic constraints have a much higher coefficient than the optimization objective terms.

Thereafter, the high-cost path points and their adjacent path points are selected to construct high-cost sub-trajectories. Then these sub-trajectories are optimized based on stochastic AGD and heuristic optimization. Moreover, if I_h is empty, the path points are sorted in descending order according to their cost value, and a certain proportion of the points with the highest cost are selected for local optimization. The principle of incremental optimization is shown in Fig. 6.

In addition to upsampling, the gradient descent technique is also indispensable in motion optimization. Ghadimi's stochastic AGD (Ghadimi & Lan, 2016) not only ensures the optimal convergence rate when solving convex problems but also can handle some non-convex

Algorithm 2. The modified motion optimization based on Ghadimi's accelerated gradient descent.

Input: θ_0 , $\{\alpha_k\}$ s.t. $\alpha_1 = 1$ and $\alpha_k \in (0,1)$ for any $k \geq 2$, $\{\beta_k > 0\}$, N_A -the number of iterations for AGD, and $\{\lambda_k > 0\}$, F_{tol} and θ_{tol} .

```

1. Initialize: Set the initial points  $\theta_0^{ag} = \theta_0$  and  $k = 1$ ,  $\alpha_k = 1$ ,  $isLocalMin = false$ ;
2. for  $k = 1$  to  $N_A$  do
3.   Set
4.      $\theta_k^{md} = (1-\alpha_k)\theta_{k-1}^{ag} + \alpha_k\theta_{k-1}$ ;
5.   Compute  $ared$ ,  $b_l$  and  $b_u$ ;
6.   if  $ared \notin (b_l, b_u)$  then
7.     reject  $\{\theta_k, \theta_k^{md}, \theta_k^{ag}\}$ ;
8.   Recalculate  $L_F$ , update  $isLocalMin$ ;
9. if  $isLocalMin$  then
10.  Optimize the trajectory base on DGA;
11. end
12.   break;
13. end
14. Compute the gradient of the objective function  $\nabla F(\theta_k^{md})$ , and update  $\alpha_k, \beta_k, \lambda_k$ ;
15. Set
16.    $\theta_k = \theta_{k-1} - \lambda_k \nabla F(\theta_k^{md})$ ;
17.    $\theta_k^{ag} = \theta_k^{md} - \beta_k \nabla F(\theta_k^{md})$ ;
18. if  $\nabla \theta_k^{md} < \theta_{tol}$  &  $\nabla F(\theta_k^{md}) < F_{tol}$  then
19.   break;
20. end
21. end
22.  $\theta_*^{md} = \theta_k^{md}$ ;
23. return  $\theta_*^{md}, \nabla F(\theta_*^{md})$ ;

```

problems. Therefore, this study applies Ghadimi's stochastic AGD to motion optimization and adds a heuristic optimization approach to this framework to avoid local optima.

Algorithm 2 shows the core iteration process of trajectory planning based on Ghadimi's AGD. In Algorithm 2, $\{\beta_k\}$ and $\{\lambda_k\}$ are the step sizes of different strategies. Both the step size strategies can be adjusted by changing λ_k and β_k . L_F is Lipschitz constant (Feng et al., 2022). Moreover, to evaluate and improve the effect of each iteration, the upper limit (b_u) and lower limit (b_l) of the actual cost reduction ($ared = F_k - F_{k-1}$) should be calculated based on the Wolfe condition (Wolfe, 1969) as follows:

$$\begin{cases} b_l = c_{low} \cdot \left(\nabla^T F_{k-1}^{md} (\theta_k^{md} - \theta_{k-1}^{md}) - \frac{L_F}{2} \|\theta_k^{md} - \theta_{k-1}^{md}\|^2 \right) \\ b_u = c_{up} \cdot \left(\nabla^T F_{k-1}^{md} (\theta_k^{md} - \theta_{k-1}^{md}) + \frac{L_F}{2} \|\theta_k^{md} - \theta_{k-1}^{md}\|^2 \right) \end{cases} \quad (17)$$

where c_{low} and c_{up} are coefficients. If $ared$ does not belong to (b_l, b_u) , L_F should be updated according to (Feng et al., 2022). If a stuck condition is found according to L_F , the planning process should be restarted based on the heuristic approach, which will be introduced in the next section.

4. Adaptive fine-tuning heuristic optimization

4.1. Directed-mutagenesis genetic algorithm based on the GA and AGP

This section proposes a modified heuristic optimization approach based on the directed-mutagenesis genetic algorithm (DGA), which involves an AGP-based probabilistic model updating mechanism. In this updating mechanism, expectation maximization and the exponential moving average method are used to estimate the GP parameters. The specific steps of AGP are as follows:

First, the cost of sampled trajectories is calculated and the first K^* trajectories with the lowest cost are reserved for the next iteration.

Second, the exponential form is adopted to calculate the probability metric of each sub-trajectory according to the cost:

$$w_F(\theta_k) = e^{\frac{h(cost_{\theta_k} - cost_{\min K})}{cost_{\max K} - cost_{\min K}}} \quad (18)$$

where h is a constant for adjusting the probability distribution, $cost_{\min K}$ and $cost_{\max K}$ are the minimum and maximum costs of the set of sampled trajectories respectively. The vector containing the above probability metrics is named $\mathbf{W}_F$. The trajectories are sorted in descending order of the weights, and the weights of the first K^* trajectories are normalized ($\mathbf{W}_n$). The mean vector μ^* and covariance matrix $\mathbf{Cov}^*$ are updated, thus the mean vector μ is also updated as follows:

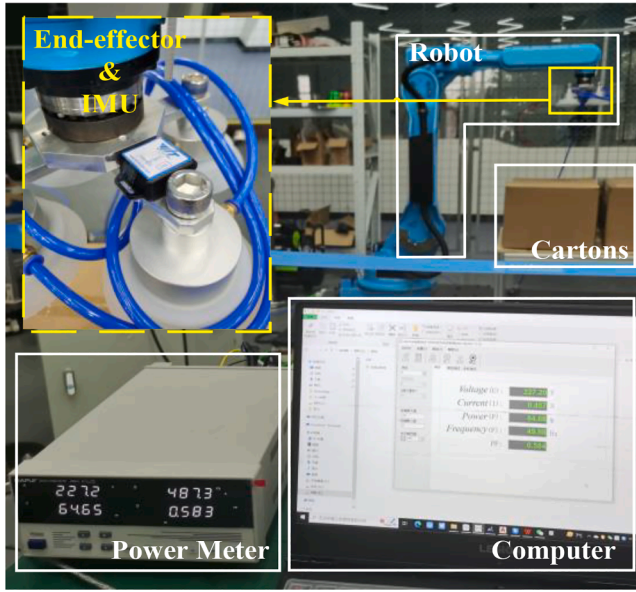


Fig. 7. The experimental equipment.

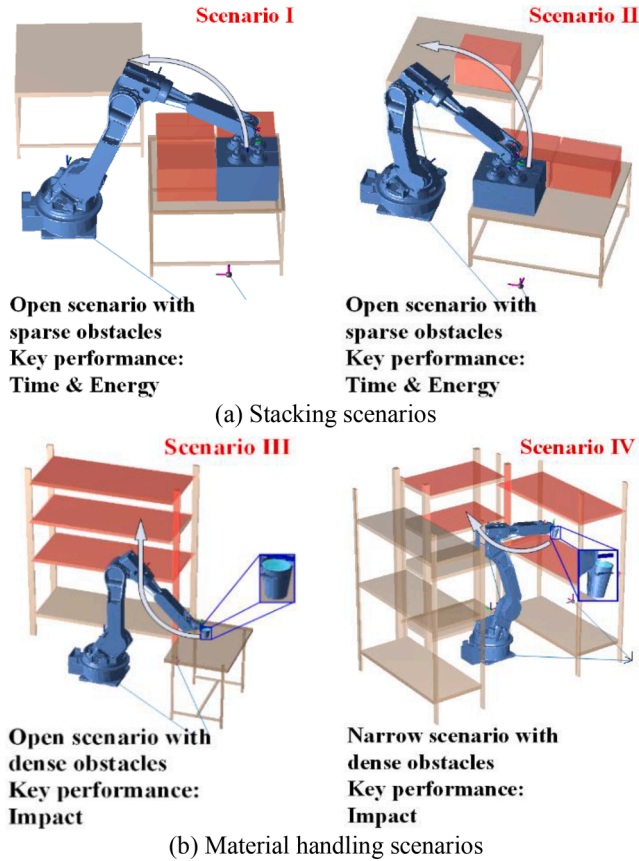


Fig. 8. The test scenarios.

Table 2

AGD-DGA configuration parameters.

Parameter	Value
w_1 in all scenarios	1.0
w_2 in scenarios I and II	7.0×10^{-3}
w_3 in scenarios I and II	3.9×10^{-4}
w_2 in scenario III	3.0×10^{-3}
w_3 in scenario III	1.0×10^{-2}
w_2 in scenario IV	8.0×10^{-3}
w_3 in scenario IV	1.8×10^{-2}
w_4 in all scenarios	28.0
w_5 in all scenarios	1.2×10^4
Path point number of θ_0	5
N_{pmax}	20
S_u	12
S_l	3
N_u	3
N_{io}	7
N_o	30
N_A	30
N_g	12
N_{s1} in scenario I, II and III	1
tol_{o1} in scenarios I, II and III	2
tol_{o2} in scenarios I, II and III	0.1
η_w in scenario I, II and III	0.5
N_{s1} in scenario IV	2
tol_{o1} in scenarios IV	0
tol_{o2} in scenarios IV	0
η_w in scenario IV	0.4
p_s	40 %
p_c	10 %
p_{nm}	10 %
p_{dm}	40 %

Table 3

Driving energy in the scenarios.

	E_m/J	E_e/J	μ
Path 1	320.85	324.53	98.85 %
Path 2	263.82	284.77	92.06 %
Path 3	278.65	295.79	93.85 %
Path 4	396.25	411.32	96.20 %
Path 5	249.85	258.17	96.67 %
Path 6	306.23	350.47	85.55 %
Path 7	220.16	239.55	91.19 %
Path 8	222.10	242.77	90.70 %
Path 9	211.58	238.03	87.50 %

Table 4

Total energy consumption in the scenarios.

	E_m/J	E_e/J	μ
Path 1	1737.3	1542.2	88.77 %
Path 2	1624.7	1473.5	90.69 %
Path 3	1526.1	1355.2	88.80 %
Path 4	2771.3	2485.4	89.68 %
Path 5	2321.0	2075.7	89.43 %
Path 6	2162.6	1936.5	89.54 %
Path 7	1477.7	1454.3	98.42 %
Path 8	1491.9	1454.6	97.50 %
Path 9	1492.1	1444.1	96.78 %

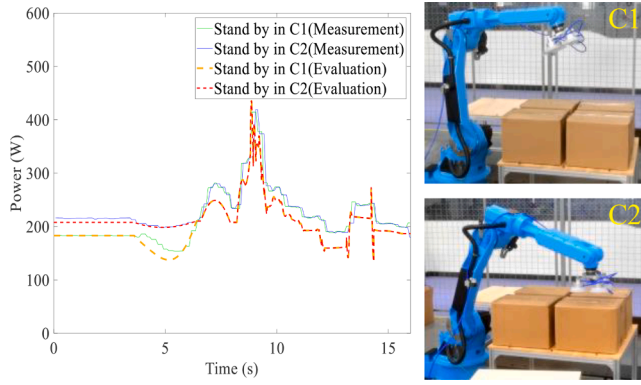


Fig. 9. Total power curves of the motion processes with different standby states, where standby state C1 has a stationary power of 183.35 W and standby state C2 has a stationary power of 216.26 W.

$$W_n = \frac{W_{F^*}}{\text{sum}(W_{F^*})} \quad (19)$$

$$\mu^* = \Theta_{K^*} \cdot W_n \quad (20)$$

$$\mu = \mu^* \quad (21)$$

$$\text{Cov}^* = (\Theta_{K^*} - \mu^*) \cdot \text{diag}(W_n) \cdot (\Theta_{K^*} - \mu^*)^T \quad (22)$$

where W_{F^*} contains the first K^* largest probability metric of W_F and Θ_{K^*} is a set consisting of the first K^* trajectories.

Third, the Cov is updated by referring to the bias correction method of Adam (Kingma and Ba, 2015):

$$\overline{\text{Cov}}_i = \begin{cases} \eta \text{Cov}_0 & \text{if } i = 0 \\ (1 - \eta) \cdot \overline{\text{Cov}}_{i-1} + \eta \cdot \text{Cov}^* & \text{if } i > 0 \end{cases} \quad (23)$$

$$\text{Cov}_i = \frac{\overline{\text{Cov}}_i}{1 - (1 - \eta)^i} \quad (24)$$

where η is the weight used to adjust Cov_i , and Cov_0 can be defined manually. In the experiments of this study, Cov_0 is defined as a diagonal matrix whose principal diagonal elements are all 2.

Fourth, if Cov_i is less than the threshold Cov_{tol} or Eq. (25) is valid, it is considered to converge and return the optimal trajectory θ_{opt} ; otherwise, if the number of iterations is greater than the upper limit, return the initial trajectory θ_{h0} .

$$\frac{F(\theta_{opt}(t)) - F(\theta_{h0}(t))}{F(\theta_{h0}(t))} \leq cFtol \quad (25)$$

where θ_{h0} is the initial trajectory of heuristic optimization, and $cFtol$ is a threshold used to determine whether heuristic optimization meets expectations.

Based on the above-described updating mechanism, the genetic algorithm is introduced into the optimization process. The specific process is shown in Algorithm 3.

In the initialization phase, the initial population is generated based on sampling and rank-based fitness assignment. In this study, Eq. (18) is also used as the fitness function.

Thereafter, the “crossover”, “selection”, and “mutation” operations are performed. Specifically, the “selection” operation is performed to retain the elite samples of each generation according to fitness, the “crossover” operation is performed to generate new low-cost samples from these elite samples, and the “mutation” operation is performed to generate random samples. It should be noted that the mutation operation is divided into “nondirected mutation” and “directed mutagenesis” in this study. Nondirected mutation randomly samples within the

feasible region, while directed mutagenesis samples based on AGP. According to the incremental optimization mechanism, the above operations are only used for high-cost path points.

The traditional mutation operation is limited by planning time and sampling scale, making it difficult to overcome local optima in a limited number of iterations. However, its randomness is the basis of probabilistic completeness. Although the sampling range of AGP is affected by μ and Cov , it has been shown to effectively improve the success rate and efficiency (Jiang, Dong, Xie, & Sweeney, 2010). Therefore, the synergy of the above operations helps to improve the success rate and performance of motion optimization.

4.2. Adaptive fine-tuning strategy for population size

Stochastic AGD can achieve the optimal convergence rate of nonlinear programming among the gradient descent-based algorithms. However, it still cannot solve the problem of local convergence. In this study, the main role of DGA is to overcome local minima, and the optimization is mainly realized through stochastic AGD. In previous works, the population size is often artificially defined, which makes it difficult to achieve a balance between the optimization effect and planning efficiency. Therefore, an adaptive population size fine-tuning method is proposed based on the path point cost gradient of adjacent iterations, which saves unnecessary calculating, sorting, and selecting operations. To maximize efficiency while maintaining performance improvement, the number of samples is dynamically adjusted based on the optimization effect of each step.

In the adaptive fine-tuning strategy, the population size is adjusted according to the trajectory cost at each step, and its range is $[S_u, S_l]$. On the premise that the population size S_p does not exceed the limit, if the cost decreases, the population size is reduced by one; otherwise, the population size is increased by one. This ensures that the population size is always at a dynamically optimal state. The fine-tuning approach is shown in Algorithm 4.

Thus, the influence of path point number on time complexity can be analyzed according to Algorithms 1 to 4. Firstly, in Algorithm 1, the incremental optimization (Algorithm 4) is regarded as the basic operation, since the N_u and N_{io} are constants, the time complexity of Algorithm 1 is $O(1)$. Secondly, in Algorithm 4, the AGD (Algorithm 2) and DGA (Algorithm 3) are the basic operations, and in the worst case, $P_o = N_p - 2$, then the time complexity of incremental optimization is $O(N_p)$. Thirdly, since the N_A and N_g are constants, the time complexity of AGD and DGA is $O(1)$ and $O(S_p)$, respectively. Therefore, the overall time complexity of AGD-DGA is $O(N_p \times S_p)$, the AFS-PS and incremental mechanism can reduce the value of S_p and P_o respectively, thus effectively improving the computational efficiency.

Since it is difficult to theoretically analyze the impact of AFS-PS on time complexity, the next section will make an analysis based on the experimental results.

5. Experiment and discussion

The computer used in the tests was equipped with a 3.6 GHz AMD Ryzen 3500X CPU and 16 GB of memory. Other hardware included a KENBEI ABD-20 industrial robot, an IMU of Wit company (for impact) and a NAPUI power meter (for energy consumption). The experimental equipment is shown in Fig. 7. Both the IMU and the power meter had appropriate kits to record the measured data in real time.

To verify the effect of the proposed AGD-DGA algorithm in different scenarios, four scenarios are designed. Fig. 8 shows the setup of the scenarios, where the obstacles in scenarios I to III are sparse, and the feasible space in scenario IV is narrow.

The simulation environment was constructed based on MATLAB (.mex files). The user interface of MATLAB is highly intuitive, and there are several toolboxes available for expert systems, such as Fuzzy Logic Toolbox and Neural Network Toolbox (Jabbar & Khan, 2016).

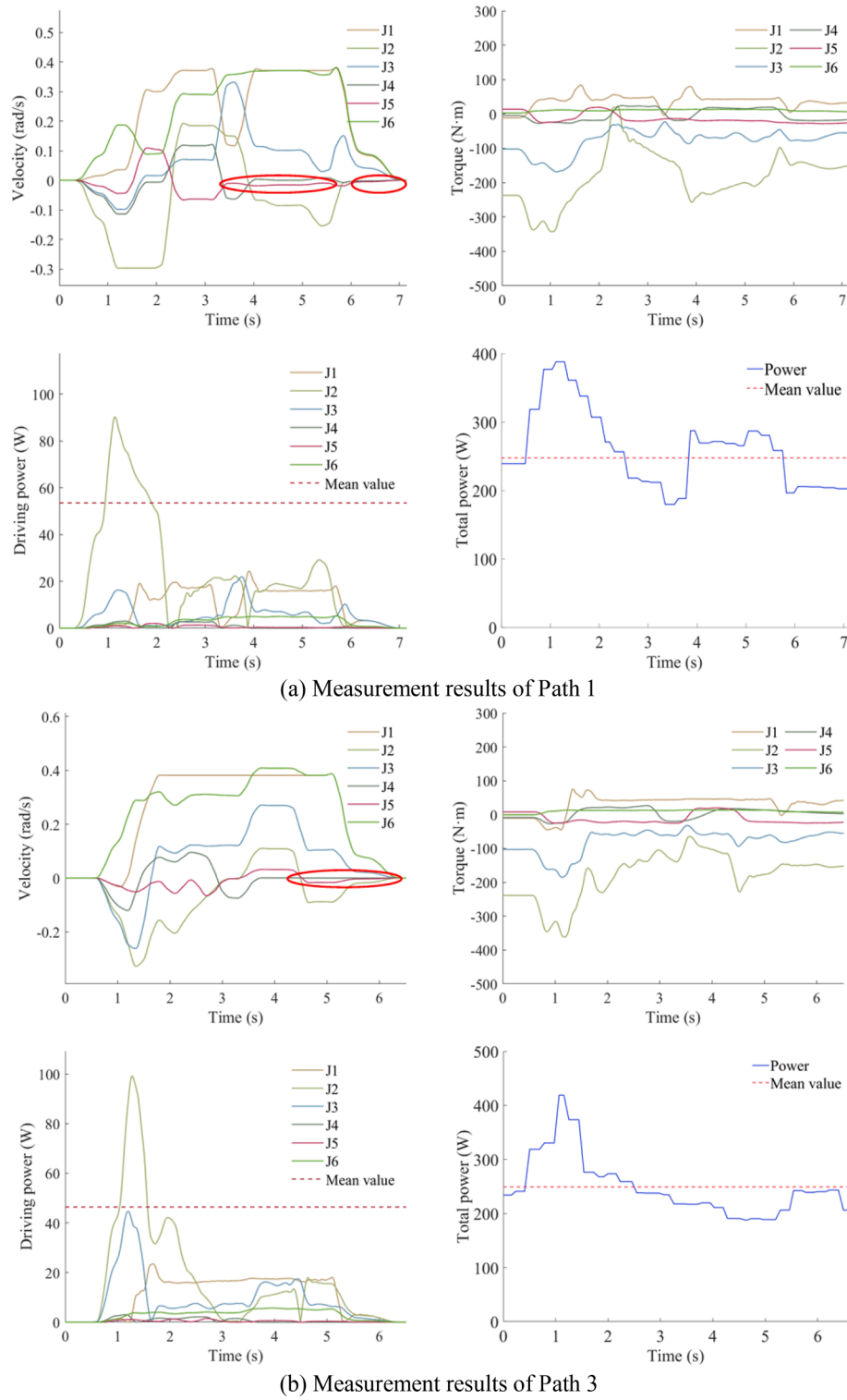


Fig. 10. Examples of measurement results, where the driving power curve shows the absolute value, and the stationary states of the joints are marked by red circles.

Table 5
Groups of experiments using different improvements.

	DGA	AFS-PS	DLM	HWU
Group (1)	–	–	–	–
Group (2)	+	–	–	–
Group (3)	+	+	–	–
Group (4)	+	+	+	–
Group (5)	+	+	+	+

Therefore, the fuzzy expert system mentioned in Section 2.1 was also constructed based on the Fuzzy Logic Toolbox of MATLAB.

The algorithms used for testing were executed 100 times in each simulation test to avoid interference caused by special factors. The statistical results included the performance indicators and calculation time of the algorithms. The motion time, energy consumption, and impact of each trajectory were estimated according to the performance evaluation model in Section 2. Then, the planning results with the minimum cost of each group were selected for the experiments to verify the actual effect

Table 6

The effect of the improvements on the proposed algorithm.

	Scenario I (Key performance: Time, energy consumption)					Scenario II (Key performance: Time, energy consumption)				
	Group (1)	Group (2)	Group (3)	Group (4)	Group (5)	Group (1)	Group (2)	Group (3)	Group (4)	Group (5)
T_c (s)	4.89	3.80	2.72	2.86	2.97	7.24	5.61	4.18	4.34	4.16
T_m (s)	8.17	8.14	7.83	6.52	6.48	10.46	10.44	10.44	9.70	9.83
E_t (J)	1742.20	1728.80	1664.60	1413.00	1407.80	2143.40	2138.80	2146.50	1993.60	2017.20
I_e (1/s ²)	9.72	9.14	9.20	7.19	7.16	7.46	7.45	7.65	8.05	8.19
Success Rate	99 %	99 %	100 %	98 %	100 %	100 %	100 %	100 %	100 %	100 %
	Scenario III (Key performance: Impact)					Scenario IV (Key performance: Impact)				
	Group (1)	Group (2)	Group (3)	Group (4)	Group (5)	Group (1)	Group (2)	Group (3)	Group (4)	Group (5)
T_c (s)	2.86	2.53	2.08	2.19	2.04	3.15	2.72	2.63	2.69	2.43
T_m (s)	6.63	6.71	6.71	6.02	6.01	5.80	5.78	5.77	5.61	5.64
E_t (J)	1472.60	1490.20	1490.20	1368.30	1364.90	1237.20	1236.70	1239.10	1211.10	1210.90
I_e (1/s ²)	7.03	6.76	6.74	6.32	6.24	8.43	6.96	7.23	5.26	4.89
Success Rate	100 %	100 %	100 %	100 %	100 %	67 %	76 %	74 %	50 %	82 %

T_c : calculation time; T_m : motion time; E_t : total energy consumption; I_e : end impact. The same below.

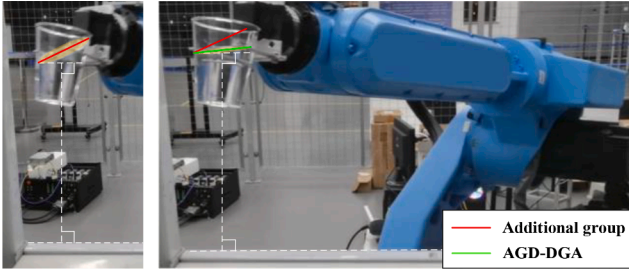


Fig. 11. Comparison of liquid sloshing amplitude in each group, the water surface is marked in yellow.

of the algorithms. It should be noted that only the algorithms of each group differ in the tests. The other experimental components such as evaluation methods, IMUs, and power meters remained the same in all tests, ensuring the validity of the conclusions.

Since the algorithms of the control groups have been able to obtain high-quality results, it is difficult for AGD-DGA to achieve a significant improvement in all performance indicators. The purpose of this study is to optimize key performance while minimizing the degradation of other performance, which will be discussed later. Table 2 shows the configuration parameters of the proposed AGD-DGA.

5.1. The effect verification of the improvement approaches

The ablation study was conducted to validate the effect of the improvement approaches (DLM, AFS-PS, HWU, and DGA) proposed in this study. Following is a detailed analysis.

First, to verify the validity of the DLM, the evaluated and measured values of energy consumption were recorded, and the accuracy of the evaluation was calculated using Eq. (30):

$$\mu = \left(1 - \left| \frac{E_m - E_e}{E_m} \right| \right) \times 100\% \quad (30)$$

In Eq. (30), μ is the evaluation accuracy of energy consumption, while E_e and E_m denote the evaluated and measured energy consumption, respectively. Tables 3 and 4 show the evaluation accuracy for driving energy and total energy consumption. The measured total energy consumption was obtained by the power meter shown in Fig. 7, and the measured values of driving energy were obtained based on the measurement results of the motor torque and velocity. The evaluation method in Table 3 is based on the dynamics model (Paryanto, Kohl, Merhof, Spreng, & Franke, 2014; Rassölkin et al., 2011), and the E_e in Table 4 is calculated based on DLM.

Furthermore, a set of comparison cases in scenario I was designed to illustrate another important role of DLM. In these cases, the cycle time was set to 16 s, and it was assumed that the robot must stand by in different configurations, and wait for other equipment to complete tasks

Table 7

Comparison of simulation results.

Scenario	Indices	CHOMP	iSAGO	GPMP2	SMTO	CDW-PSO	AGD-DGA
Scenario I (Key performance: Time, energy consumption)	T_c (s)	4.76	3.68	4.53	10.42	65.96	2.97
	T_m (s)	7.36	8.02	7.65	9.63	12.83	6.48
	E_t (J)	1763.20	1714.80	1635.90	2034.3	2698.7	1407.80
	I_e (1/s ²)	13.83	10.14	9.22	13.20	12.61	7.16
	Success Rate	99 %	99 %	100 %	99 %	100 %	100 %
Scenario II (Key performance: Time, energy consumption)	T_c (s)	9.69	5.23	6.43	11.39	74.17	4.16
	T_m (s)	12.23	10.60	10.61	10.89	14.01	9.83
	E_t (J)	2620.60	2179.20	2168.10	2235.6	2966.4	2017.20
	I_e (1/s ²)	11.31	7.38	6.59	9.33	12.07	8.19
	Success Rate	96 %	100 %	100 %	99 %	100 %	100 %
Scenario III (Key performance: Impact)	T_c (s)	3.71	2.37	2.69	4.62	23.99	2.04
	T_m (s)	6.11	6.61	6.70	6.74	11.54	6.01
	E_t (J)	1371.40	1470.40	1485.60	1533.3	2481.3	1364.90
	I_e (1/s ²)	7.72	6.99	6.87	8.06	6.72	6.24
	Success Rate	100 %	100 %	100 %	100 %	100 %	100 %
Scenario IV (Key performance: Impact)	T_c (s)	3.57	2.61	2.99	4.56	26.58	2.43
	T_m (s)	5.92	5.81	5.82	6.31	13.09	5.64
	E_t (J)	1270.30	1239.00	1238.30	1364.7	2833.7	1210.90
	I_e (1/s ²)	7.42	8.72	9.44	11.27	9.79	4.89
	Success Rate	62 %	68 %	59 %	64 %	97 %	82 %

Table 8

Comparison of experimental results.

Scenario	Indices	CHOMP	iSAGO	GPMP2	SMT0	CDW-PSO	AGD-DGA
Scenario I (Key performance: Time, energy consumption)	T_m (s)	6.68	6.42	6.19	6.80	8.10	5.94
	E_t (J)	1751.0	1648.7	1529.6	1742.0	1967.9	1513.6
	I_e (1/s ²)	14.98	8.35	10.64	13.13	19.90	9.48
Scenario II (Key performance: Time, energy consumption)	T_m (s)	11.22	9.93	10.09	9.28	10.09	8.86
	E_t (J)	2733.6	2315.9	2173.9	2141.9	2303.6	2131.8
	I_e (1/s ²)	12.05	8.94	8.20	13.68	13.59	9.48
Scenario III (Key performance: Impact)	T_m (s)	5.93	5.72	6.15	6.23	7.01	5.90
	E_t (J)	1469.9	1455.9	1467.6	1539.3	1666.6	1463.8
	I_e (1/s ²)	7.03	6.89	6.29	6.35	8.63	5.75
Scenario IV (Key performance: Impact)	T_m (s)	5.33	5.84	5.70	5.76	8.85	5.58
	E_t (J)	1278.9	1429.2	1345.4	1355.9	2165.2	1303.1
	I_e (1/s ²)	6.58	5.96	7.52	7.58	10.07	5.43

Note: For each group, the results with the best key performance are selected for the experiments.

before stacking ($C1: \theta_{C1} = [-0.28, -0.09, 0.03, 0.00, 1.26, -0.28]^T$, $C2: \theta_{C2} = [-0.28, 0.37, -0.06, 0.00, 1.26, -0.28]^T$). The total power curves of different processes are shown in Fig. 9.

Fig. 9 shows that the power at stationary (power loss) accounts for most of the total power. According to the measurement and calculation results, the total energy consumption of the C2 configuration is 3683.2 J; while that of the C1 configuration is lower (3471.8 J) due to its lower power, despite requiring additional movements. Meanwhile, the calculated energy consumption evaluation accuracies of C1 and C2 are 92.3 % and 92.8 %, respectively. According to the above data, DLM not only helps to improve the accuracy of energy consumption evaluation but also plays a guiding role in the setting of the static configuration.

According to the above tables and figures, the proposed energy consumption model maintains an accuracy of more than 85 % in all test paths, and the average accuracy is above 90 %. Another noteworthy phenomenon is that the driving energy is not always positively correlated with total energy consumption, and the difference between the two is large, meaning that the total energy consumption cannot be accurately estimated through the dynamics model alone during optimization. The main reasons are as follows:

- (1) Fig. 9 shows that the power at stationary cannot be ignored (200 W), which is significantly greater than the additional energy consumed during motion.
- (2) Traditional methods cannot calculate the energy consumption of stationary joints at certain moments, as marked by the red circles in Fig. 10.

Although the driving energy consumption of Path 3 are higher than that of Path 2, the actual total energy consumption of Path 3 is lower because of its lower energy loss. If the traditional method is used and only driving energy consumption is considered, the opposite conclusion will be drawn, which demonstrates that performance modeling affects the performance.

Moreover, the impact evaluation accuracy is calculated according to the same method, the average accuracy of all paths is higher than 80 %, indicating that the impact assessment method is also effective.

Thereafter, the effect of each improvement measure is also verified through a comparative test. Table 5 presents the test group settings in the ablation study, where the algorithm of Group (5) is the proposed AGD-DGA, and the energy consumption of Groups (1) to (3) is evaluated based on the dynamics model (Paryanto, Kohl, Merhof, Spreng, & Franke, 2014; Rassölkin et al., 2011). The simulation test results are shown in Table 6.

In this study, DGA and HWU are proposed to improve the success rate. The data in Table 6 verifies that the DGA improves the success rate of optimization. In narrow scenario IV, the success rate is increased by 9 % because of DGA. Moreover, the HWU can also effectively improve the success rate of planning in narrow scenes, where the additional performance indices significantly impact success rates (see Groups (3) and (4)

of Scenario 4). The HWU effectively addressed this problem by dynamically adjusting the weight of the secondary priority, which improves the success rate by 32 %, without impacting motion performance in scenes with sparse obstacles.

The goal of AFS-PS is to improve planning efficiency on the premise of rarely affecting optimization results. In all four scenarios, the average planning time is reduced by up to 28.4 % because of AFS-PS, while the average performance remains roughly the same. According to the principle of AGD-DGA, trajectory optimization is mainly realized through the AGD, and DGA is only used to overcome the local minimum problem. Therefore, appropriate adjustment of the population size does not significantly affect the performance of the planning results. Moreover, due to the improvement of the probabilistic model, DGA also reduces the planning time by 11.5 % to 22.5 %.

DLM accurately evaluates the components of energy consumption, and as a result, Group (4) reduced motion time by up to 16.7 % and energy consumption by up to 15.1 % compared to Group (3) based on the dynamics model. Thus, the energy consumption evaluation method does affect the energy consumption. In addition, according to Group (3) and Group (4), DLM does not significantly increase the computation time. This is because DLM only increases a cubic polynomial, which has acceptable time complexity.

Scenario III focuses on the smoothness of the end motion. In this scenario, an additional group that does not consider impact was set and tested. The average impact of the additional group is 6.58/s², and the impact of Group (5) is reduced by 5.2 % compared to the additional group. Moreover, the impact indicators can also be evaluated intuitively by observing the liquid sloshing. The solutions of the two groups with the lowest impact cost in scenario III were selected for experimental testing. The liquid slosh amplitude of each group during the liquid handling process was observed and analyzed through videos. Fig. 11 shows that the liquid slosh amplitude of Group (5) is significantly smaller than that of the additional group. Overall, the data and figures validate the effectiveness of impact optimization.

5.2. Baseline comparison

To fully reflect the effectiveness of the proposed algorithm, it was compared with different types of state-of-the-art algorithms, including iSAGO (Feng et al., 2022), GPMP2 (Mukadam et al., 2018), CHOMP (Zucker et al., 2013), STMO (Osa, 2020) and CDW-PSO (Yang et al., 2022). All control groups performed comprehensive motion performance optimization with the same motion performance weights as the AGD-DGA, and all groups used a global velocity of 12 % of the maximum velocity. The simulation results are shown in Table 7.

It can be seen from Table 7 that AGD-DGA achieves better optimization results on the whole. Specifically, the calculation time is significantly reduced, and the motion performance indices are improved.

First, in terms of planning efficiency, AGD-DGA achieves higher efficiency, and its planning time is 6.9 % to 95.5 % less than that of other

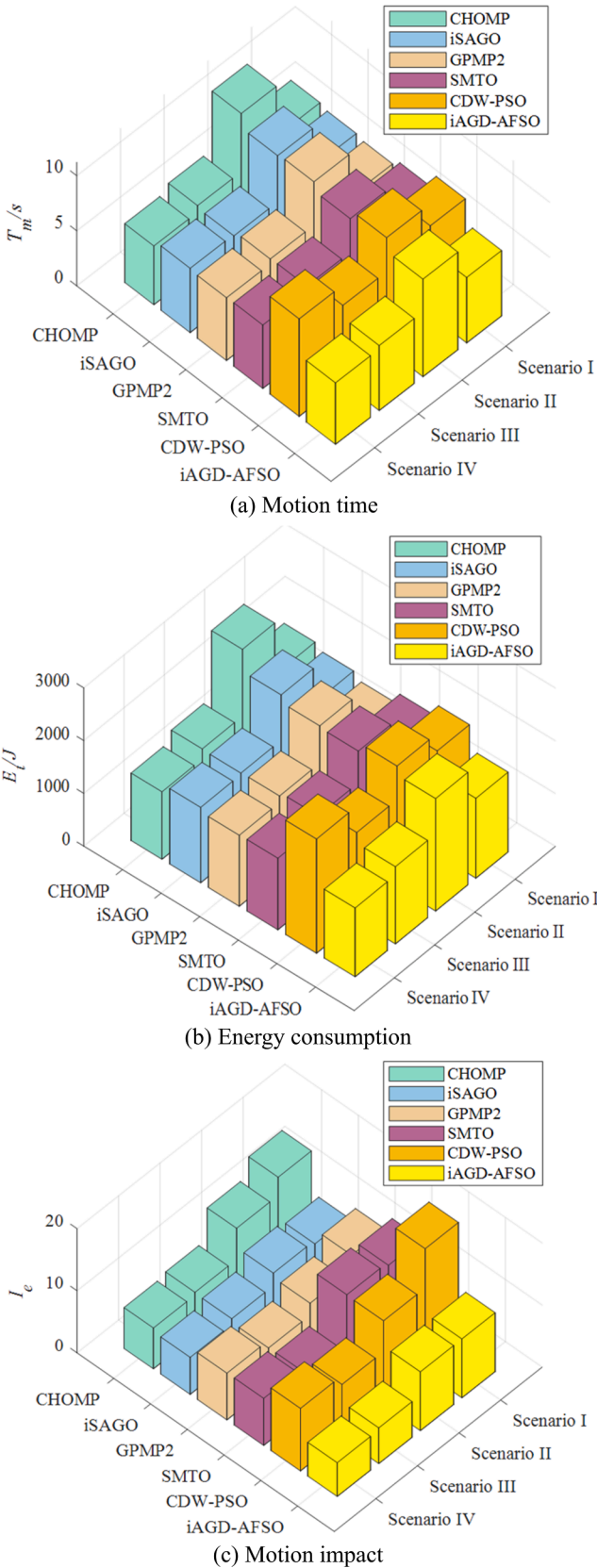


Fig. 12. Comparison of the performance indices of different algorithms in the experiments.

algorithms. On the one hand, the DGA improves the sampling quality through the adaptive Gaussian process, which accelerates convergence. On the other hand, the AFS-PS dynamically adjusts the population size to avoid redundant sampling, evaluation, and sorting.

Second, in terms of motion performance, DLM and DGA effectively improve the key performance. In the stacking scenarios, the motion time is reduced by 7.3 % to 56.9 %, and the energy consumption is reduced by 7.0 % to 57.3 % compared with the control groups. Moreover, in terms of end impact, in the liquid handling scenarios, the impact is reduced by 7.1 % to 56.6 %.

Third, in scenario IV, the narrow feasible space and additional performance indicators pose a challenge to the planning success rate. Under this premise, compared with most control algorithms (CHOMP, iSAGO, GPMP2, SMT0), AGD-DGA has an advantage in terms of success rate, which increases by 14 % to 23 %. Since the success rate in narrow scenarios is sensitive to cost calculation, HWU effectively improves the success rate by adjusting the weight of the secondary priority in the early stage. Furthermore, the improvement of DGA in the optimization process also has a positive impact on the success rate.

In addition, according to Table 7, CDW-PSO achieves the highest success rate. However, as a heuristic algorithm, CDW-PSO cannot effectively utilize gradient information, and it is difficult to converge stably and efficiently to the optimal solution within a limited number of iterations, so it lags behind other algorithms significantly in terms of motion performance and solving efficiency.

To verify the effectiveness of the simulation, the optimal trajectories of each group were selected for the experiment. The equipment shown in Fig. 7 accurately quantified the motion performance of the planning results. The results are shown in Table 8 and Fig. 12, which validate that the AGD-DGA optimizes the key performance.

In scenarios I and II, the robot performed the stacking tasks, thus the motion time and the energy consumption were given priority. The motion time of AGD-DGA is reduced by 4.0 % to 26.7 %, and the energy consumption is reduced by up to 23.1 % compared with other groups.

In scenarios III and IV, the robot performs the liquid handling tasks. To avoid liquid splashing, the impact of the end-effector was given priority. Table 8 shows that the impact is reduced by 8.6 % to 46.1 %, and it also performed well in terms of time and energy consumption.

Fig. 13 shows the motion process of each scenario. The above results prove that AGD-DGA can not only meet the requirements of obstacle avoidance but also achieve better performance.

6. Conclusions and future work

In this paper, a motion optimization method called AGD-DGA is proposed. The experimental results demonstrate the characteristics of the proposed method, which are summarized as follows:

- (1) The energy consumption and end impact are included in the optimization objectives to improve the comprehensive motion performance. In particular, a modified energy consumption model called DLM is proposed to address the evaluation accuracy problem of existing methods. The DLM considers both driving power and power loss, resulting in an average accuracy above 90 % in all test scenarios, which ensures the effectiveness of the optimization results. Meanwhile, the adoption of DLM also played a positive role in motion performance, reducing energy consumption by up to 15.1 % in the tests. Moreover, the motion time was reduced by up to 16.7 % and the end impact was reduced by 5.2 %, demonstrating the effectiveness of the optimization for the T-E-I metrics.
- (2) To improve the performance of optimization results and the success rate of gradient-based optimization under limited iterations, a modified heuristic optimization approach named DGA is proposed, which includes a novel directed mutagenesis operation based on the adaptive Gaussian process. According to the test



Fig. 13. The photos present the robot movement of the experimental group (AGD-DGA) in the real machine test. The curves in the figure show the trajectories of the end-effector, where the curves of CHOMP are red, the curves of AGD-DGA are green, the curves of iSAGO are blue, and the curves of GPMP2 are magenta, the obstacles are highlighted in red, and the cup is highlighted in cyan.

Algorithm 3. DGA-based motion optimization.

Input: θ_{ho} , N_g -the maximum number of iterations for genetic algorithm, p_s -the selection probability, p_c -the crossover probability, p_{dm} -the directed mutagenesis probability, p_{nm} -the nondirected mutation probability, S_p -population size, Cov_{tol}

1. **Initialize:** Generate initial population G_{init} ;
2. Calculate the cost of all individuals and obtain elite individuals;
3. Update the probabilistic model based on AGP;
4. **for** $i = 1$ to N_g **do**
5. **Nondirected mutation:** Randomly generate n_{nm} individuals, $n_{nm} = \text{round}(p_{nm} \cdot S_p)$;
6. **Directed mutagenesis:** Generated n_{dm} individuals based on the probabilistic model, $n_{dm} = \text{round}(p_{dm} \cdot S_p)$;
7. **Selection:** Retain n_s elite individuals to the next generation, $n_s = \text{round}(p_s \cdot S_p)$;
8. **Crossover:** Generated n_c individuals based on the elite individuals, $n_c = S_p - n_{nm} - n_{dm} - n_s$;
9. Calculate the cost of all individuals and obtain elite individuals;
10. Update the probabilistic model based on AGP;
11. **if** Eq. (25) is true **or** $Cov_i < Cov_{tol}$ **then**
12. **break**;
13. **end**
14. **end**
15. **return** optimal individual θ_{opt} as θ_{opt}^{md} ;

Algorithm 4. Incremental optimization with AFS-PS.

Input: θ , P_{hc} -The set of path points with high-cost value, S_u , S_b , F_{tol} , θ_{tol} , tol_{o2} -Threshold of obstacle cost of θ_{*}^{md} , N_o -The number of iterations for every high-cost path point, P_o -The number of path points to optimize (in AGD-DGA, P_o is the size of P_{hc})

```

1. Initialize: set  $isLocalMin = false$ ,  $S_p = S_u$ ;
2. Reinitialize  $\{\theta_0^{md}, \theta_0^{ag}\}$  based on  $\theta$ ;
3. for  $i = 1$  to  $P_o$  do
4.   for  $j = 1$  to  $N_o$  do
5.     Solve  $\theta_{*}^{md}$  and  $\nabla F(\theta_{*}^{md})$  based on Algorithm 2 and Algorithm 3;
6.     if  $\nabla \theta_{*}^{md} < \theta_{tol}$  &  $\nabla F(\theta_{*}^{md}) < F_{tol}$  then
7.        $\theta = \theta_{*}^{md}$ ;
8.       return  $\theta$ ;
9.     end
10.  end
11.  $\theta = \theta_{*}^{md}$ ;
12. if the cost of  $\theta$  goes down &  $S_p > S_l$  &  $f_{obs}(\theta) \leq tol_{o2}$  then
13.    $S_p = S_p - 1$ ;
14. elseif cost of  $\theta$  goes up &  $S_p < S_p$ 
15.    $S_p = S_p + 1$ ;
16. end
17. end
18. return  $\theta$ ;

```

results, DGA increased the success rate by 9 % and improved the computing efficiency by 11.5 % to 22.5 %.

- (3) Considering the effect of an inappropriate sampling scale on the efficiency of random sampling, an adaptive fine-tuning strategy named AFS-PS is proposed, which can dynamically adjust the population size of DGA. In ablation studies, AFS-PS can reduce the calculation time by 28.4 %.

In summary, the experimental results validate that the proposed AGD-DGA can achieve higher planning efficiency and better motion performance than the existing mainstream algorithms, which is beneficial to engineering tasks requiring high repeatability, low energy consumption, and low impact. The proposed algorithm can also be integrated into the expert systems for path planning to help decision-making in industrial scenarios. Despite its effectiveness, the proposed AGD-DGA still cannot achieve true stable convergence due to the complexity of optimization objectives and search space, resulting in the randomness of planning results in real scenarios, and this limitation will be a focus of our future research.

CRediT authorship contribution statement

Shengjie Li: Conceptualization, Methodology, Software, Writing – original draft. **Jin Wang:** Resources, Writing – review & editing, Supervision. **Haiyun Zhang:** Writing – review & editing. **Yichang Feng:** Software. **Guodong Lu:** Resources, Supervision. **Anbang Zhai:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

The work of this paper is supported by the Key R&D Program of Zhejiang Province (No. 2020C01025), the National Natural Science Foundation of China (No. 52175032), “Pioneer” and “Leading Goose” R&D Program of Zhejiang (No.2022C01101, 2023C01070), Yuyao Science and technology project (No.2023JH03010019), and Robotics Institute of Zhejiang University Grant (K12107, K11808). Moreover, we would like to thank MogoEdit (<https://www.mogoedit.com>) for its English editing during the preparation of this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2023.122794>.

References

- Burger, B., Maffettone, P. M., Gusev, V. V., Aitchison, C. M., Bai, Y., Wang, X., ... Cooper, A. I. (2020). A mobile robotic chemist. *Nature*, 583(7815). <https://doi.org/10.1038/s41586-020-2442-2>
- Cao, X., Yan, H., Huang, Z., Ai, S., Xu, Y., Fu, R., & Zou, X. (2021). A multi-objective particle swarm optimization for trajectory planning of fruit picking manipulator. *Agronomy*, 11(11). <https://doi.org/10.3390/agronomy11112286>
- Do, C. B., & Batzoglu, S. (2008). What is the expectation maximization algorithm? *Nature Biotechnology*, 26(8). <https://doi.org/10.1038/nbt1406>
- Dong, J., Mukadam, M., Boots, B., & Dellaert, F. (2018). Sparse Gaussian processes on matrix lie groups: A unified framework for optimizing continuous-time trajectories.

- Proceedings - IEEE International Conference on Robotics and Automation. <https://doi.org/10.1109/ICRA.2018.8461077>
- Du, X., Yu, T., & Chen, K. (2023). An orthodontic path planning method based on improved gray wolf optimization algorithm. *Soft Computing*. <https://doi.org/10.1007/s00500-023-08924-0>
- Ekrem, Ö., & Aksoy, B. (2023). Trajectory planning for a 6-axis robotic arm with particle swarm optimization algorithm. *Engineering Applications of Artificial Intelligence*, 122. <https://doi.org/10.1016/j.engappai.2023.106099>
- Fang, T., & Ding, Y. (2022). A sampling-based motion planning method for active visual measurement with an industrial robot. *Robotics and Computer-Integrated Manufacturing*, 76. <https://doi.org/10.1016/j.rcim.2022.102322>
- Feng, Y., Wang, J., Zhang, H., & Lu, G. (2022). Incrementally stochastic and accelerated gradient information mixed optimization for manipulator motion planning. *IEEE Robotics and Automation Letters*, 7(4). <https://doi.org/10.1109/LRA.2022.3191206>
- Ghadimi, S., & Lan, G. (2016). Accelerated gradient methods for nonconvex nonlinear and stochastic programming. *Mathematical Programming*, 156(1–2), 59–99. <https://doi.org/10.1007/s10107-015-0871-8>
- Ghorbanpour, A., & Richter, H. (2022). Energy-optimal impedance control of cooperative robot manipulators. *Journal of Dynamic Systems, Measurement and Control, Transactions of the ASME*, 144(12). <https://doi.org/10.1115/1.4055391>
- Henkel, C., & Toussaint, M. (2020). Optimized directed roadmap graph for multi-agent path finding using stochastic gradient descent. Proceedings of the ACM Symposium on Applied Computing. <https://doi.org/10.1145/3341105.3373916>
- Jabbar, H. K., & Khan, R. Z. (2016). Tools of development of expert systems: A comparative study. Proceedings of the 10th INDIACOM; 2016 3rd International Conference on Computing for Sustainable Global Development, INDIACOM 2016.
- Jiang, X., Dong, B., Xie, L., & Sweeney, L. (2010). Adaptive gaussian process for short-term wind speed forecasting. *Frontiers in Artificial Intelligence and Applications*, 215. <https://doi.org/10.3233/978-1-60750-606-5-661>
- Kalakrishnan, M., Chitta, S., Theodorou, E., Pastor, P., & Schaal, S. (2011). STOMP: Stochastic trajectory optimization for motion planning. *IEEE International Conference on Robotics and Automation, ICRA, 2011*, 9–13. <http://library1.nida.ac.th/termpaper/6/sd/2554/19755.pdf>
- Kang, M., Shin, H., Kim, D., & Yoon, S. E. (2020). TORM: Fast and accurate trajectory optimization of redundant manipulator given an end-effector path. *IEEE International Conference on Intelligent Robots and Systems*. <https://doi.org/10.1109/IROS45743.2020.9341358>
- Karnopp, D. (1985). Computer simulation of stick-slip friction in mechanical dynamic systems. *Journal of Dynamic Systems, Measurement and Control, Transactions of the ASME*, 107(1). <https://doi.org/10.1115/1.3140698>
- Kayastha, S., Katupitiya, J., Pearce, G., & Rao, A. (2023). Comparative study of post-impact motion control of a flexible arm space robot. *European Journal of Control*, 69. <https://doi.org/10.1016/j.ejcon.2022.100738>
- Kingma, D. P., & Ba, J. L. (2015). Adam: A method for stochastic optimization. 3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings.
- Li, Y., Wang, Z., Yang, H., Zhang, H., & Wei, Y. (2023). Energy-optimal planning of robot trajectory based on dynamics. *Arabian Journal for Science and Engineering*, 48(3). <https://doi.org/10.1007/s13369-022-07185-7>
- Liu, J., Wang, H., Li, X., Chen, K., & Li, C. (2023). Robotic arm trajectory optimization based on multiverse algorithm. *Mathematical Biosciences and Engineering*, 20(2). <https://doi.org/10.3934/mbe.2023130>
- Mak, D. K. (2021). Exponential Moving Average.. https://doi.org/10.1007/978-3-030-70622-7_4
- Mukadam, M., Dong, J., Yan, X., Dellaert, F., & Boots, B. (2018). Continuous-time Gaussian process motion planning via probabilistic inference. *International Journal of Robotics Research*, 37(11), 1319–1340. <https://doi.org/10.1177/0278364918790369>
- Noreen, I., Khan, A., Asghar, K., & Habib, Z. (2019). A path-planning performance comparison of RRT*-AB with MEA* in a 2-Dimensional Environment. *Symmetry*, 11(7). <https://doi.org/10.3390/sym11070945>
- Osa, T. (2020). Multimodal trajectory optimization for motion planning. *International Journal of Robotics Research*, 39(8). <https://doi.org/10.1177/0278364920918296>
- Paryanto, B., Kohl, J., Merhof, J., Spreng, S., & Franke, J. (2014). Energy consumption and dynamic behavior analysis of a six-axis industrial robot in an assembly system. *Procedia CIRP*, 23(C), 131–136. <https://doi.org/10.1016/j.procir.2014.10.091>
- Pastras, G., Fysikopoulos, A., & Chrysosouris, G. (2019). A theoretical investigation on the potential energy savings by optimization of the robotic motion profiles. *Robotics and Computer-Integrated Manufacturing*, 58. <https://doi.org/10.1016/j.rcim.2019.02.001>
- Paulus, I. (2014). Ganzheitliche Betrachtung – Energie- und CO2-Reduzierung in der Herstellung der Karosserie. https://doi.org/10.1007/978-3-658-05980-4_2
- Rassolkhan, A., Höimoja, H., & Teemets, R. (2011). Energy saving possibilities in the industrial robot IRB 1600 control. 2011 7th International Conference-Workshop Compatibility and Power Electronics, CPE 2011 - Conference Proceedings, 1, 226–229. <https://doi.org/10.1109/CPE.2011.5942236>
- Safeea, M., Béarée, R., & Neto, P. (2020). Collision avoidance of redundant robotic manipulators using newton's method. *Journal of Intelligent and Robotic Systems: Theory and Applications*, 99(3–4), 673–681. <https://doi.org/10.1007/s10846-020-01159-3>
- Singh, G., & Banga, V. K. (2022). Kinematics and trajectory planning analysis based on hybrid optimization algorithms for an industrial robotic manipulators. *Soft Computing*, 26(21), 11339–11372. <https://doi.org/10.1007/s00500-022-07423-y>
- Wolfe, P. (1969). Convergence conditions for ascent methods. *SIAM Review*, 11(2). <https://doi.org/10.1137/1011036>
- Yang, Y., & Xu, H. (2022). Time-optimal trajectory optimization of serial robotic manipulator with kinematic and dynamic limits based on improved particle swarm optimization. *International Journal of Advanced Manufacturing Technology*, 120(1–2). <https://doi.org/10.1007/s00170-022-08796-y>
- Yi, J., Yuan, Q., Sun, R., & Bai, H. (2022). Path planning of a manipulator based on an improved P_RRT* algorithm. *Complex and Intelligent Systems*, 8(3), 2227–2245. <https://doi.org/10.1007/s40747-021-00628-y>
- Yu, X., Dong, M., & Yin, W. (2022). Time-optimal trajectory planning of manipulator with simultaneously searching the optimal path. *Computer Communications*, 181, 446–453. <https://doi.org/10.1016/j.comcom.2021.10.005>
- Zhang, B., Wang, L., & Wu, J. (2019). Dynamic parameters and friction model identification of an industrial hybrid robot. ASME International Mechanical Engineering Congress and Exposition, Proceedings (IMECE), 4. <https://doi.org/10.1115/IMECE2019-10916>
- Zhang, J., Zhang, J., Zhang, Q., & Wei, X. (2022). Obstacle avoidance path planning of space robot based on improved particle swarm optimization. *Symmetry*, 14(5). <https://doi.org/10.3390/sym14050938>
- Zhang, T., Zhang, M., & Zou, Y. (2021). Time-optimal and smooth trajectory planning for robot manipulators. *International Journal of Control, Automation and Systems*, 19(1), 521–531. <https://doi.org/10.1007/s12555-019-0703-3>
- Zhong, J., Wang, T., & Cheng, L. (2022). Collision-free path planning for welding manipulator via hybrid algorithm of deep reinforcement learning and inverse kinematics. *Complex and Intelligent Systems*, 8(3). <https://doi.org/10.1007/s40747-021-00366-1>
- Zucker, M., Ratliff, N., Dragan, A. D., Pivtoraiko, M., Klingensmith, M., Dellin, C. M., ... Srinivasa, S. S. (2013). CHOMP: Covariant Hamiltonian optimization for motion planning. *International Journal of Robotics Research*, 32(9–10), 1164–1193. <https://doi.org/10.1177/0278364913488805>